

Patch, that is, drones that can autonomously inspect, identify and repair potholes in roads.

The third is Fire and Forget robots, which can operate perpetually within live utility pipes, handling tasks like inspection, repair, metering and reporting.

The multi-million-pound project is actually deeper than it appears to be. It will involve not just advanced research in drones and robotics, but also include development of advanced simulators like airflow simulators to study air pollution, tackle the problem of aging water pipelines and more. The project will also study the environmental and social impact of having a robot workforce always on duty in the city, trying to achieve the task with minimal disturbance to the city's dwellers.

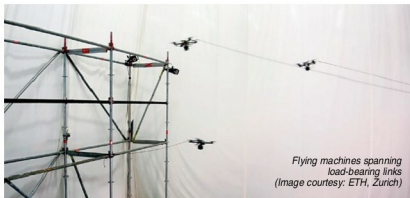
A trick or two learnt from flies and birds

Think small, flying, fast and agile, and insects are the first creatures that come to mind. How challenging it is to vanquish a fly that irritatingly hovers around you, buzzing into your ears! So close by, yet so tough to swat. Well, it is no surprise then that with similar goals, drone researchers are turning to nature for inspiration.

Insects are able to fly swiftly in and out of even the trickiest of spaces without colliding into anything. This is a capability that UAVs must have, if these are to work autonomously and coexist with humans. One way to achieve this would be to

More exciting applications...

- ETH Zurich recently demonstrated drones building a rope bridge as part of their Aerial Construction Project. The researchers navigated quadcopters between two sets of scaffolding. Thereafter, the drones were able to survey the distance, figure out how to build the bridge and with the help of motorised spools, tied together ropes to make a 7.3-metre (24-feet)-long bridge, without any human intervention.



- Robot scientists from across Europe have joined hands under the auspices of Aerial Robotics Cooperative Assembly System (ARCAS) project to develop drones that can fly in a coordinated way to share the weight of heavy building materials. This will help deliver heavy payloads to difficult areas not reachable by cranes, which often pose risks to human workers.
- Drones are expected to be very helpful for farmers, too. Stevia, the sugar substitute maker, considered deploying drones with lights over their farms to promote crop growth at night, too. A contest held last year in Maryland, USA, challenged students to develop drones that could prevent infestation of corn crops. Some interesting designs involved drones that landed on infested crops and pulled out insects with mechanical arms.
- NASA has shown how UAVs can be used to hunt hurricanes, while Lockheed Martin showed how these can help in detecting and predicting avalanches, volcanic eruptions, wildfires and other natural disasters.
- Conservation Drones is an organisation focused on using drones for environmental-conservation activities such as protecting chimpanzees in Tanzania and tracking Sumatran orangutans.
- IBM is developing a drone called IRIS+, which can play table tennis. IRIS+ can automatically track the trajectory of a ball coming to its side of the table and return the ball to the other player.

equip the drones with digital cameras that capture a 360-degree image of what is around them. However, this goes against the form factor and weight requirements, which are

critical for drones.

So scientists at Swiss Federal Institute of Technology (ETH) Zurich, Switzerland, have now developed an insect-inspired motion sensor



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